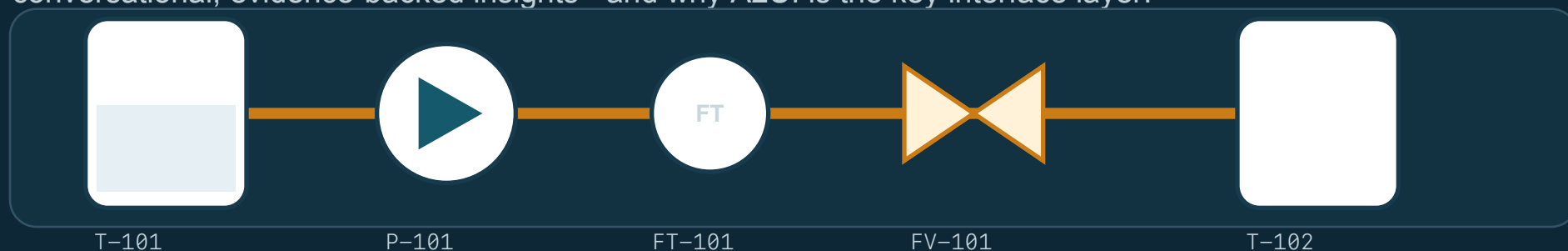


DEMO ENABLEMENT

Industrial operations assistant Workflow + A2UI guide

A presenter-ready explanation of how process diagrams and synthetic DCS data become conversational, evidence-backed insights - and why A2UI is the key interface layer.



2 inputs

P&ID + historian export

2,166

synthetic DCS readings

10

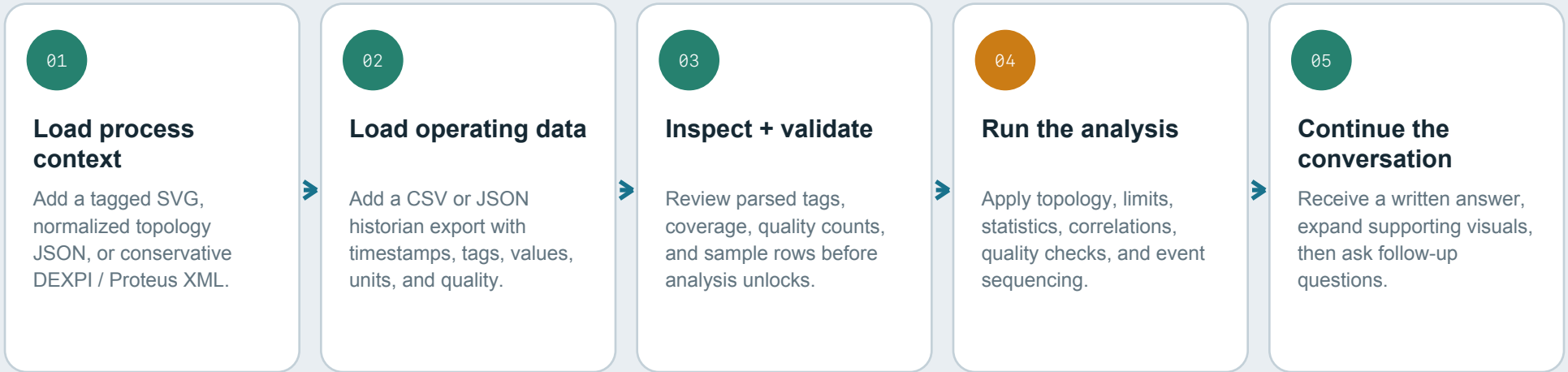
approved A2UI components

0 API keys

required for the default demo

From source files to an explainable conversation

The demo has a visible beginning, middle, and end: evidence enters, the system validates it, and the assistant turns it into a discussion with attached visual proof.



DEMO DATA AT A GLANCE

Process path

T-101 > P-101 > FT-101 > FV-101 > T-102
Provenance: open DEXPI reference specimen retained under CC BY 4.0; the working T-101 topology and DCS readings are synthetic.

Historian

6 tags | 1-minute cadence | 08:00-14:00

Quality

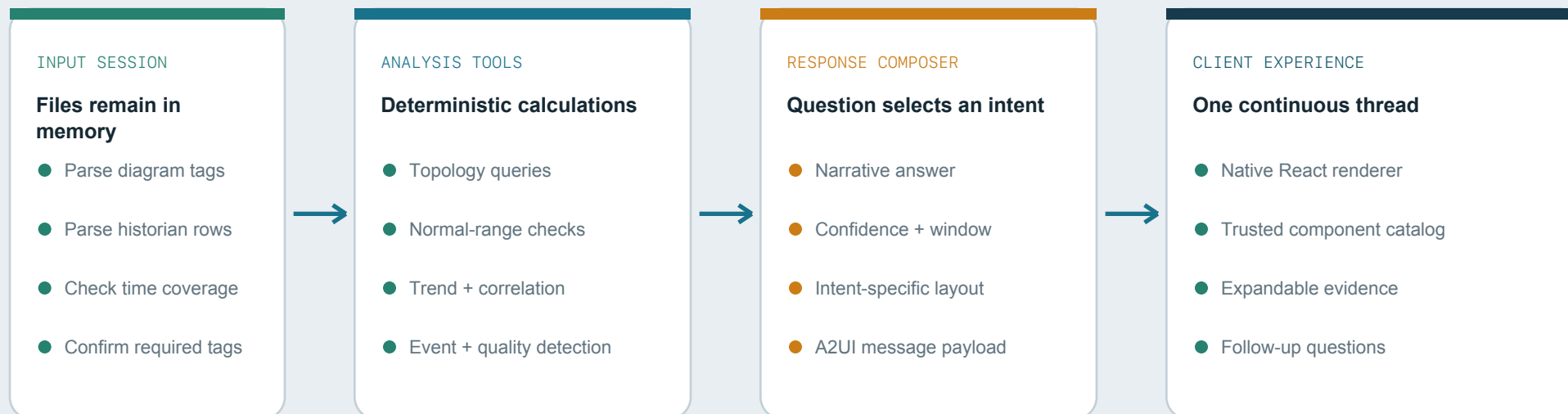
2,140 GOOD | 20 BAD | 6 MISSING

SAY The prototype does not jump straight to an answer. It lets the user see exactly what went in, what was parsed, and when the system is ready.

03 SECTION 1 - WORKFLOW

What the prototype is actually doing

A local, repeatable evidence path keeps the demonstration reliable while preserving the same boundaries needed for a future agent runtime.



DESIGNED INCIDENT

10:15-11:00: flow falls to 28.7 m3/h, pressure rises to 5.93 bar, and LT-101 climbs at 31.4%/h. The flow-pressure correlation is -0.985, which is consistent with a downstream restriction signature.

WHAT IS PROVEN

The readings, time window, limits, topology, and calculated relationships are traceable to the supplied demo inputs.

WHAT IS NOT CLAIMED

Correlation does not prove a mechanical cause. The UI says 'consistent with' and requires field confirmation.

A six-minute live demo runbook

Use the sample path for reliability, then slow down at the input inspection and A2UI evidence moments.

TIME	MOMENT	WHAT TO DO	WHAT THE TEAM SEES
0:00	Inputs	Choose Load sample inputs	Both sources become visible and analysis unlocks.
0:45	Inspection	Open Inspect on each file	Show diagram tags, 2,166 rows, time coverage, and quality.
1:30	General review	Choose Analyze these inputs	The assistant explains the recovered state at 14:00.
2:30	Root cause	Ask: Why did T-101's level increase?	Narrative + KPIs + trends + timeline + highlighted path.
4:15	Data quality	Ask: Which sensor readings are unreliable?	The layout changes to a quality warning and tag table.
5:15	Close	Expand Supporting evidence	Call out A2UI, the trusted catalog, and advisory boundaries.

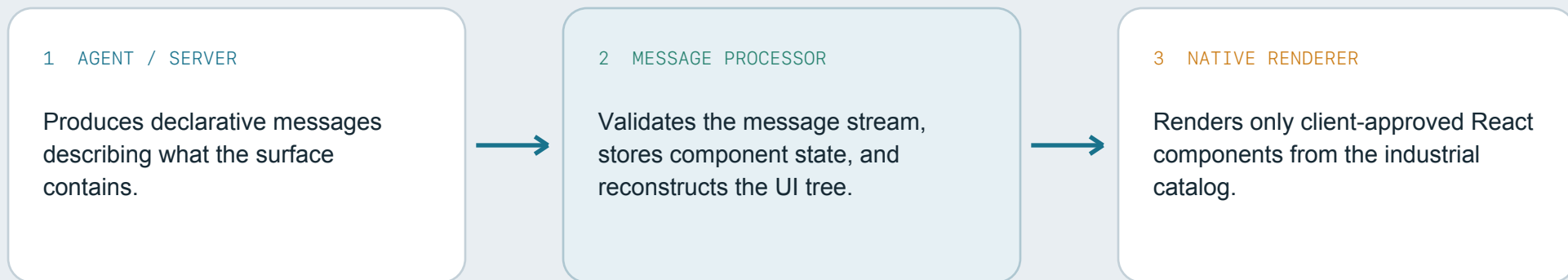
SAY

Watch how the interface changes with the question. The assistant is not merely describing a chart - it is choosing an evidence layout appropriate to the operator's intent.

05 SECTION 2 - A2UI

A2UI in one sentence

A2UI is a JSON-based protocol that lets an agent describe a user interface while the client retains control of the components, design system, and rendering behavior.



THE THREE MESSAGES USED IN FIELDGUIDE

`createSurface`

Open a named rendering surface and identify the approved catalog.

`updateComponents`

Send the component list: summary, KPIs, chart, timeline, diagram, and more.

`updateDataModel`

Attach question, intent, and provenance data to the surface state.

THE IMPORTANT SAFETY PROPERTY

The server sends data and component descriptions - not arbitrary JavaScript or executable UI code.

PROTOTYPE-SPECIFIC NOTE

A2UI supports streaming. This demo returns the three-message bundle in one local HTTP response for repeatability.

06 SECTION 2 - A2UI

One question, one adaptive response surface

The narrative remains conversational. The A2UI surface underneath it carries the visual evidence selected for that intent.

SERVER RESPONSE / SIMPLIFIED

```
{ "version": "v0.9",  
  "createSurface": {  
    "surfaceId": "process-insight-...",  
    "catalogId": ".../industrial/v1"  
  }  
}  
  
{ "updateComponents": {  
  "components": [  
    { "id": "root",  
      "component": "ResponseLayout",  
      "children": ["kpis", "trend",  
        "timeline", "diagram", "inspection"] },  
    { "id": "trend",  
      "component": "TrendChart", ... }  
  ]  
}  
  
{ "updateDataModel": {  
  "value": { "intent": "root_cause" }  
}}
```

CLIENT-RENDERED EVIDENCE

A downstream restriction is most likely

Written answer + confidence + incident window

LT-101

78.7 %

PT-101

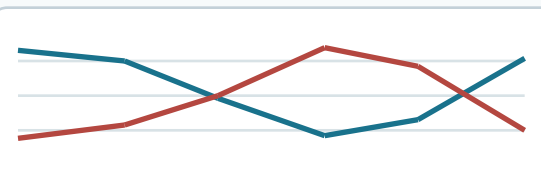
5.93 bar

FT-101

28.7

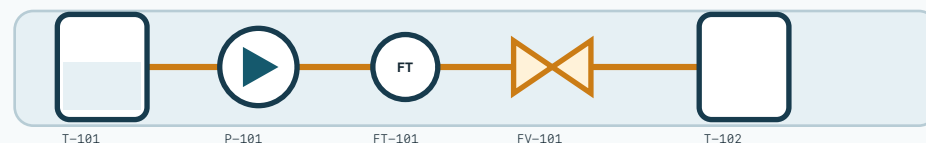
FV-101

82 %



TIMELINE

- Flow low
- Pressure high
- Level rising



1

Surface + catalog establish the contract.

2

Root child IDs define the response composition.

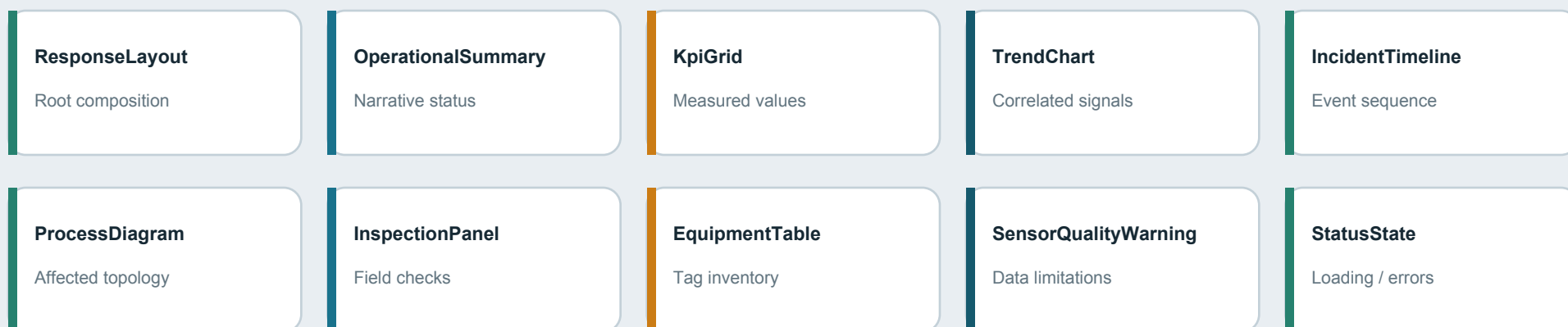
3

The renderer turns approved types into native UI.

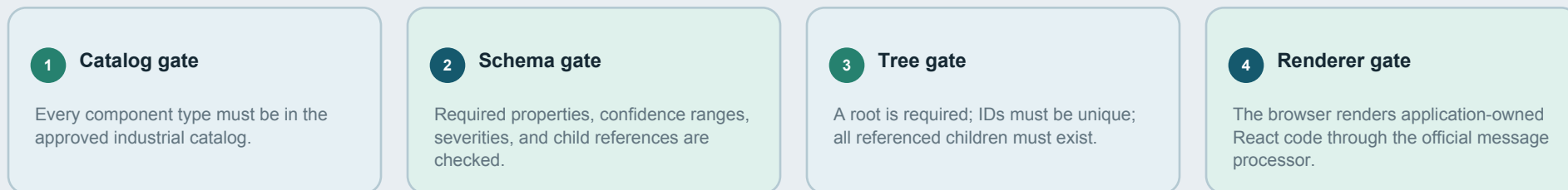
07 SECTION 2 – A2UI

The restricted catalog is the control point

FieldGuide does not let the server invent arbitrary widgets. The client and server share one industrial catalog with ten approved component types.



VALIDATION BEFORE RENDER



SAY A2UI moves the flexibility boundary: the server can choose the composition, but the client still owns which components exist and how they behave.

08 SECTION 2 – A2UI

Why this matters beyond the proof of concept

A2UI gives the team a path from a reliable demo to richer agent experiences without surrendering interface governance.

TEXT-ONLY ASSISTANT

The user must interpret prose, find the relevant trend, connect the tags, and decide what to inspect.

One answer shape for every question

A2UI EVIDENCE ASSISTANT

The response combines plain language with the right native visual surface: KPIs for status, trends for correlation, a timeline for sequence, a P&ID; for topology, or a quality warning for unreliable data.

The layout adapts while design and safety controls remain stable

BUSINESS + PRODUCT BENEFITS

1

Adaptive

Question intent changes the evidence layout.

2

Governed

Only approved component schemas can render.

3

Explainable

Narrative, values, time windows, and visuals stay attached.

4

Consistent

The client's design system controls the final experience.

5

Portable

The protocol can target web, mobile, and desktop renderers.

6

Extensible

Add approved components without rewriting the whole agent.

Closing message

FieldGuide is not just a chatbot on top of historian data. It is a controlled method for turning engineering context and operating evidence into an interface the user can inspect, question, and trust.